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GTC Law Group PC & Affiliates
One University Ave., Ste. 302B
Westwood, MA 02090

EXAMINER	
ALGIBHAH, HAMZA N	

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2441	

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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/123,195	Applicant(s) Fang et al.	
	Examiner HAMZA N ALGIBHAH	Art Unit 2441	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 03/17/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-29 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-7,9-23 and 25-29 is/are rejected.

8) ☑ Claim(s) 8 and 24 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 03/17/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Detailed Action

1. ***Claims 1-29*** are pending.

2. ***Claims 1-7, 9-23 and 25-29 are rejected.***

3. ***Claims 8 and 24 are objected.***

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

4. ***Claims 1-7 and 19-23 are rejected*** under 35 U.S.C. 102(a)(2) as being anticipated by Selvarajan et al (Pub. No.: US 2018/0357561 A1).

As per claim 1, Selvarajan discloses an apparatus, comprising:

- a parameter acquisition circuit structured to interpret a plurality of vehicle parameter values (Selvarajan, Fig 1B paragraph 0031, wherein a vehicle electronic data broker gateway receives data from data sources wherein the received data can be the plurality of vehicle parameter values and the component within the gateway that receive the data can be the acquisition circuit as claimed); and

- a parameter conditioning circuit structured to condition the plurality of vehicle parameter values for storage in one or more cache devices (Selvarajan, Fig 1B paragraph 0031, wherein the data conversion unit, as the conditioning circuit, is operative to convert data fetched (stored in a data cache) from the data source into a standard format supported by the

APIs}; ***and***

- ***a parameter storage circuit structured to store the conditioned plurality of vehicle parameter values in the one or more cache devices*** {Selvarajan, Fig 1B, 3, paragraph 0031, wherein data broker unit 140 also includes at least one data cache 172, which is operative to store the data received from the data source before transmitting the data to the client}***;***

As pre claim 2, claim 1 is incorporated and Selvarajan discloses **wherein at least one of the one or more cache devices comprises a non-volatile memory** {Selvarajan, Fig 1B, 3, paragraph 0061, wherein suitable processor-readable media may include storage or memory media such as magnetic or optical media. For example, storage or memory media may include conventional hard disks, compact discs, DVDs, Blu-ray discs, or other optical storage media; volatile or non-volatile media such as Random Access Memory (RAM); Read Only Memory (ROM), Electrically Erasable Programmable ROM (EEPROM), flash memory, and the like; or any other media that can be used to carry or store desired program code in the form of computer executable instructions or data structures};

As pre claim 3, claim 2 is incorporated and Selvarajan discloses **wherein the parameter storage circuit is further structured to store the conditioned plurality of vehicle parameter values in the non-volatile memory in response to a data communication limitation** {Selvarajan, abstract, paragraph 0050-0051, wherein the data broker gateway can temporarily store the data that is being sent to a client in the data cache. If the same data gets requested by a different client, and if the data is fresh enough, the data broker gateway can send the same data to this new client as well. This reduces the traffic in the bus, reduces the interrupt to the server, and reduces the latency time as well};

As pre claim 4, claim 1 is incorporated and Selvarajan discloses **wherein the parameter conditioning circuit is further structured to determine a storage location value; the parameter storage circuit is further structured to store the conditioned plurality of vehicle parameter values in response to the storage location value; and the one or more cache devices are disposed onboard a vehicle** (Selvarajan, abstract, paragraph 0056, wherein the gateway can be hosted as a certified application, such as on a flight deck 714 of an aircraft and thus the cache is disposed onboard a vehicle as claimed. Since data is received from data sources and converted and stored in the cache, the gateway by inherence determine the location of the cache (onboard cache) to store the receive data);

As pre claim 5, claim 4 is incorporated and Selvarajan discloses **wherein each of the one or more cache devices disposed onboard the vehicle is associated with a controller that is distinct from controllers associated with the other of the one or more cache devices** (Selvarajan, Fig 1B, 3, paragraph 0031, wherein data broker unit 140 also includes at least one data cache 172, which is operative to store the data received from the data source before transmitting the data to the client);

As pre claim 6, claim 1 is incorporated and Selvarajan discloses **wherein: the parameter conditioning circuit is further structured to determine a storage location value; the parameter storage circuit is further structured to store the conditioned plurality of vehicle parameter values in response to the storage location value; and the one or more cache devices are disposed offboard a vehicle** (Selvarajan, abstract, paragraph 0057, wherein the gateway can be hosted as an uncertified application, off a flight deck 814 of an aircraft and thus the cache is

disposed offboard a vehicle as claimed. Since data is received from data sources and converted and stored in the cache, the gateway by inherence determine the location of the cache (offboard cache) to store the receive data};

As pre claim 7, claim 6 is incorporated and Selvarajan discloses wherein the one or more cache devices are based at least in part on a network cloud-based storage system {Selvarajan, abstract, paragraph 0057, wherein ionics data broker (ADB) gateway 810 is hosted as an uncertified application, off a flight deck 814 of an aircraft. The ADB gateway 810 operatively communicates with an onboard certified wireless gateway 820, such as through wireless communications (e.g., Wi-Fi), or with an EFB device. The wireless gateway 820 or EFB device is in operative communication with a plurality of certified applications, hosted in onboard avionics units 830 (e.g., FMS, D&G, FCS, IRS, GPS, CMF), through an avionics data bus 840. The ADB gateway 810 is also in operative wireless communications (e.g., Wi-Fi) or wired communications (via an EFB device) with uncertified applications that reside on outside data sources/clients, such carry-on mobile devices or the web};

Claims 19-23 are rejected under the same rationale as claims 1-7.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention

and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

5. ***Claim***(s) **9-18 and 25-29** is/are rejected under 35 U.S.C. 103 as being unpatentable over Selvarajan et al (Pub. No.: US 2018/0357561 A1) in view of Petousis et al (Pub. No.: US 2018/0261020 A1).

As pre claim 9, claim 1 is incorporated and Selvarajan does not explicitly disclose **wherein the parameter conditioning circuit is further structured to generate an expiration value for the plurality of vehicle parameter values structured to trigger a selective expiration of the plurality of vehicle parameter values in the one or more cache devices.** However, Petousis discloses **wherein the parameter conditioning circuit is further structured to generate an expiration value for the plurality of vehicle parameter values structured to trigger a selective expiration of the plurality of vehicle parameter values in the one or more cache devices** (Petousis, paragraph 0043, 0051-0052 wherein the header of the message is passed to the scheduling module, while the message itself is stored in short- or long-term storage. The scheduling module then schedules the message (e.g., using an optimization process, a cost-function process, etc.) based on the information in the header (e.g., data importance, message size). The headers can then be stored in short- or long-term storage (e.g., the same or different from the message storage) until the associated message is sent (e.g., upon when the headers are removed from storage or otherwise processed). When the message is to be sent (e.g., according to the schedule determined by the scheduling module), the message is retrieved from the message storage for transmission. Headers and/or messages can be moved from short-term storage to long-term storage in response to: short-term storage overflow, timer expiration, a threshold number of attempts being made, or occurrence of any other suitable event).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify Selvarajan in view of Petousis such that an expiration value is used as claimed because this would have provided a way to determine which data to keep and for how long and which data to remove based on the importance of the data and thus improve the efficiency of the system.

As pre claim 10, claim 9 is incorporated and Petousis further discloses wherein the parameter storage circuit is further structured to transmit the expiration value to the one or more cache devices (Petousis, paragraph 0043, 0051-0052 wherein the header of the message is passed to the scheduling module, while the message itself is stored in short- or long-term storage. The scheduling module then schedules the message (e.g., using an optimization process, a cost-function process, etc.) based on the information in the header (e.g., data importance, message size). The headers can then be stored in short- or long-term storage (e.g., the same or different from the message storage) until the associated message is sent (e.g., upon when the headers are removed from storage or otherwise processed). When the message is to be sent (e.g., according to the schedule determined by the scheduling module), the message is retrieved from the message storage for transmission. Headers and/or messages can be moved from short-term storage to long-term storage in response to: short-term storage overflow, timer expiration, a threshold number of attempts being made, or occurrence of any other suitable event).

As pre claim 11, claim 9 is incorporated and Petousis further discloses wherein the parameter storage circuit is further structured to selectively expire the plurality of vehicle

parameter values responsive to the expiration value (Petousis, paragraph 0049 wherein expired vehicle sensor data may be removed).

As pre claim 12, claim 9 is incorporated and Petousis further discloses wherein the expiration value is a time value corresponding to a period of time to store the plurality of vehicle parameter values in the one or more cache devices (Petousis, paragraph 0043, 0051-0052 wherein the header of the message is passed to the scheduling module, while the message itself is stored in short- or long-term storage. The scheduling module then schedules the message (e.g., using an optimization process, a cost-function process, etc.) based on the information in the header (e.g., data importance, message size). The headers can then be stored in short- or long-term storage (e.g., the same or different from the message storage) until the associated message is sent (e.g., upon when the headers are removed from storage or otherwise processed). When the message is to be sent (e.g., according to the schedule determined by the scheduling module), the message is retrieved from the message storage for transmission. Headers and/or messages can be moved from short-term storage to long-term storage in response to: short-term storage overflow, timer expiration, a threshold number of attempts being made, or occurrence of any other suitable event).

As pre claim 13, claim 9 is incorporated and Petousis further discloses comprising: a policy acquisition circuit structured to interpret a vehicle policy data value comprising at least a portion of a vehicle policy; wherein the parameter conditioning circuit is further structured to generate the expiration value responsive to the vehicle policy data value (Petousis, paragraph 0043, 0051-0052 wherein the header of the message is passed to the scheduling module, while the message itself is stored in short- or long-term storage. The scheduling module

then schedules the message (e.g., using an optimization process, a cost-function process, etc.) based on the information in the header (e.g., data importance, message size). The headers can then be stored in short- or long-term storage (e.g., the same or different from the message storage) until the associated message is sent (e.g., upon when the headers are removed from storage or otherwise processed). When the message is to be sent (e.g., according to the schedule determined by the scheduling module), the message is retrieved from the message storage for transmission. Headers and/or messages can be moved from short-term storage to long-term storage in response to: short-term storage overflow, timer expiration, a threshold number of attempts being made, or occurrence of any other suitable event).

As pre claim 14, claim 9 is incorporated and Petousis further discloses **wherein the parameter conditioning circuit is further structured to: determine a type value of the plurality of vehicle parameter values; and generate the expiration value responsive to the type value** (Petousis, paragraph 0043, 0050-0052 wherein vehicle sensor data can be grouped in three divisions (e.g., critical, operations, and applications). In this example, a fraction (e.g., within a range from 0 to a value B, where B is less than or equal to 1) of the bandwidth is reserved for each division, and a dedicated scheduling module corresponds to each division. Determining transmission order functions to determine which buffer vehicle sensor data should be transferred to. There are preferably multiple buffers, such that determining transmission order includes discriminating between different buffers, but there can alternatively be a single buffer. Determining the transmission order can optionally include determining the queuing order in which vehicle sensor data is transferred to the buffer and/or is ordered within the buffer. Determining transmission order can be based on prioritization (e.g., higher priority vehicle sensor data can be designated for transmission first, and thus transferred to a buffer designated

for high priority transmission and/or transferred to the transformation module directly), and can additionally or alternatively be based on network quality (e.g., the highest priority vehicle sensor data that can be transmitted over the available network bandwidth is transmitted first), or determined based on any other suitable parameter).

As pre claim 15, claim 14 is incorporated and Petousis further discloses wherein the type value corresponds to at least one of engine data; control data; mission critical data; motive status data; or motive operational data (Petousis, paragraph 0043, 0050-0052 wherein vehicle sensor data can be grouped in three divisions (e.g., critical, operations, and applications). In this example, a fraction (e.g., within a range from 0 to a value B, where B is less than or equal to 1) of the bandwidth is reserved for each division, and a dedicated scheduling module corresponds to each division. Determining transmission order functions to determine which buffer vehicle sensor data should be transferred to. There are preferably multiple buffers, such that determining transmission order includes discriminating between different buffers, but there can alternatively be a single buffer. Determining the transmission order can optionally include determining the queuing order in which vehicle sensor data is transferred to the buffer and/or is ordered within the buffer. Determining transmission order can be based on prioritization (e.g., higher priority vehicle sensor data can be designated for transmission first, and thus transferred to a buffer designated for high priority transmission and/or transferred to the transformation module directly), and can additionally or alternatively be based on network quality (e.g., the highest priority vehicle sensor data that can be transmitted over the available network bandwidth is transmitted first), or determined based on any other suitable parameter).

As pre claim 16, claim 14 is incorporated and Petousis further discloses wherein the type value corresponds to at least one of a vehicle state value; a vehicle mode value; a diagnostic value; or a fault value (Petousis, paragraph 0043, 0050-0052 wherein vehicle sensor data can be grouped in three divisions (e.g., critical, operations, and applications). In this example, a fraction (e.g., within a range from 0 to a value B, where B is less than or equal to 1) of the bandwidth is reserved for each division, and a dedicated scheduling module corresponds to each division. Determining transmission order functions to determine which buffer vehicle sensor data should be transferred to. There are preferably multiple buffers, such that determining transmission order includes discriminating between different buffers, but there can alternatively be a single buffer. Determining the transmission order can optionally include determining the queuing order in which vehicle sensor data is transferred to the buffer and/or is ordered within the buffer. Determining transmission order can be based on prioritization (e.g., higher priority vehicle sensor data can be designated for transmission first, and thus transferred to a buffer designated for high priority transmission and/or transferred to the transformation module directly), and can additionally or alternatively be based on network quality (e.g., the highest priority vehicle sensor data that can be transmitted over the available network bandwidth is transmitted first), or determined based on any other suitable parameter).

As pre claim 17, claim 1 is incorporated and Selvarajan does not explicitly disclose wherein the parameter conditioning circuit is further structured to condition the plurality of vehicle parameter values via compressing the plurality of vehicle parameter values. However, Petousis discloses wherein the parameter conditioning circuit is further structured to condition the plurality of vehicle parameter values via compressing the plurality of vehicle parameter values (Petousis, paragraph 0035, wherein Prioritizing the vehicle sensor data can optionally

include estimating the compressibility of the vehicle sensor data. Estimating the compressibility of the vehicle sensor data functions to estimate the minimum size of the vehicle sensor data (e.g., after compression), which can be used for scheduling the vehicle sensor data to be transformed (e.g., compressed) and transmitted).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify Selvarajan in view of Petousis such that the plurality of vehicle parameter values are compressed as claimed because this would have provided a way to reduce the size of the data which improves the efficiency of the system.

As pre claim 18, claim 1 is incorporated and Selvarajan does not explicitly disclose wherein the parameter conditioning circuit is further structured to condition the plurality of vehicle parameter values via summarizing the plurality of x chicle parameter values. However, Petousis discloses wherein the parameter conditioning circuit is further structured to condition the plurality of vehicle parameter values via summarizing the plurality of x chicle parameter values (Petousis, paragraph 0035, wherein Prioritizing the vehicle sensor data can optionally include estimating the compressibility of the vehicle sensor data. Estimating the compressibility of the vehicle sensor data functions to estimate the minimum size of the vehicle sensor data (e.g., after compression), which can be used for scheduling the vehicle sensor data to be transformed (e.g., compressed) and transmitted).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify Selvarajan in view of Petousis such that the plurality of vehicle parameter values are summarized as claimed because this would have provided a way to reduce the size of the data which improves the efficiency of the system.

Claims 25-29 are rejected under the same rationale as claims 9-18.

Allowable Subject Matter

Claims 8 and 24 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HAMZA N ALGIBHAH whose telephone number is (571)270-7212. The examiner can normally be reached 7:30 am - 3:30 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Wing Chan can be reached on (571) 272-7493. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or

571-272-1000.

/HAMZA N ALGIBHAH/
Primary Examiner, Art Unit 2441